MSSD Project Report 1

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I hereby declare that I wrote this report on my own and without the use of any other than the cited sources and tools and all explanations that I copied directly or in their sense are marked as such.

1 Introduction

1.1 Physical Problem and Motivation

The aim of this report is to simulate the Schrödinger equation as a generalized eigenvalue problem on a two-dimensional bounded domain $\Omega \subset \mathbb{R}^2$. This problem provides a non-trivial PDE eigenvalue problem whose analytic solution is known, enabling direct verification of the FEM discretization. Further this problem allows spectral time evolution of the resulting eigenbasis, which can be used to explore Lieb–Robinson (LR) bounds. LR bounds describe a phenomenon in quantum mechanics that quantum information propagates at a finite effective velocity even in non-relativistic systems [10]. The intention is to model a continuous space using a finite number of modes with a non-relativistic quantum particle governed by the Schrödinger equation.

LR bounds are a very useful concept in quantum systems and have recently been applied to quantum simulation allowing not only the classical trotterization discretization in time but due to the bound on information also a spatial LR discretization [6]. (Discretization is not completely accurate but for the sake of understanding as this is not a paper for quantum computing).

1.2 Strong Form

The goal is to find the energy eigenvalues $E \in \mathbb{R}$ and eigenfunctions $\psi : \Omega \to \mathbb{R}$ satisfying the stationary Schrödinger equation in natural units ($\hbar = 1$, $m = 1$) for a free particle confined in an infinite potential well [9]:

$$\hat{H}\psi := -\frac{1}{2}\nabla^2\psi + V(\mathbf{x})\,\psi = E\,\psi \qquad \text{in } \Omega. \tag{1}$$

The "particle in a box" or "infinite quantum well" is a popular problem that consists of a particle trapped in a domain due to an infinite potential energy V outside of the domain. Inside the domain:

$$V(\mathbf{x}) = 0, \qquad \forall\, \mathbf{x} \in \Omega. \tag{2}$$

1.3 Boundary Decomposition

The boundaries $\partial\Omega$ are the four sides of the square or the entire circumference of a circle. A homogeneous Dirichlet condition is imposed on the entire boundary:

$$\Gamma_D = \partial\Omega, \qquad \psi\big|_{\Gamma_D} = 0. \tag{3}$$

Physically, this enforces a hard-wall, indicating that the probability of finding the quantum particle outside or precisely at the geometric perimeter of the domain is zero. By preventing any quantum leakage or transmission out of the domain, this condition ensures perfect wave reflection at the boundaries. This idealized boundary allows the observation and analysis of reflections in the domain.

1.4 Weak Form

To derive the continuous weak form, the trial and test space is defined as $V_0 = H_0^1(\Omega)$. Multiplying the strong form in (1) by a test function $v \in V_0$ and integrating over the domain yields:

$$\int_\Omega \frac{1}{2}\nabla\psi \cdot \nabla v + V(\mathbf{x})\,\psi v d\mathbf{x} = E \int_\Omega \psi\, v d\mathbf{x}. \tag{4}$$

Because the test space strictly enforces $v|_{\Gamma_D} = 0$, the boundary integral vanishes identically. The resulting weak formulation reads: find the eigenpair $(\psi, E) \in V_0 \times \mathbb{R}$ such that:

$$a(\psi, v) = E\, m(\psi, v) \qquad \forall\, v \in V_0, \tag{5}$$

where the symmetric bilinear forms representing the system stiffness and mass are given by:

$$a(\psi, v) = \int_\Omega \frac{1}{2}\nabla\psi \cdot \nabla v + V(\mathbf{x})\,\psi v d\mathbf{x}, \tag{6}$$

$$m(\psi, v) = \int_\Omega \psi\, v d\mathbf{x}. \tag{7}$$

Equation (5) represents a continuous generalized symmetric eigenvalue problem. Once discretized, it can be expressed in matrix form as $A\mathbf{u} = E\,M\mathbf{u}$, where A and M are the assembled stiffness and mass matrices, respectively. Both global sparse matrices are symmetric positive definite.

1.5 Finite Element Space and Polynomial Order

The spatial domain Ω is discretized using a uniform triangular mesh. The finite element space is composed of $\mathcal{P}_1$ elements with strongly enforced homogeneous Dirichlet conditions:

$$V_h = \left\{ v_h \in C^0(\overline{\Omega}) \;\middle|\; v_h|_K \in \mathcal{P}_1(K)\ \forall\, K \in \mathcal{T}_h,\ v_h|_{\partial\Omega} = 0 \right\}, \tag{8}$$

The resulting discrete system with N elements per side has $\mathcal{O}(N^2)$ global degrees of freedom.

1.6 Simplifications and Non-dimensionalization

Some physical assumptions are made for simplicity. Setting $\hbar = m = 1$ removes all physical scaling prefactors from (1). Spatial lengths are scaled directly by the box width L, and energies are expressed in inverse squared length units proportional to the fundamental domain scale. Additionally, for simplicity there are no external magnetic fields or complex vector potentials and therefore the Hamiltonian operator is real and symmetric. The eigenfunctions can therefore be chosen purely real, reducing the complex Hermitian problem to a real symmetric system. Finally, the particle considered is spinless and non-relativistic.

1.6.1 Time Dependence via the Spectral Theorem

While the finite element method discretizes the spatial domain Ω, for the time domain an analytical approach is chosen. Since the Hamiltonian is assumed to be symmetric and real-valued, the Spectral Theorem can be applied, which guarantees a complete, orthogonal basis of spatial eigenstates $\{\phi_n\}_{n=1}^{D}$, with D the total number of degrees of freedom. This simplifies the formal time-evolution into a sum of the eigenstates multiplied with the energy eigenenvalue time evolution operator [4]:

$$\psi(\mathbf{x}, t) = e^{-i\hat{H}t}\psi_0(\mathbf{x}) = \sum_{n=1}^{D} c_n e^{-iE_n t}\phi_n(\mathbf{x}). \tag{9}$$

Physically speaking each energy component acts as a stationary state: its spatial shape and probability density remain constant over time, while its quantum phase rotates in the complex plane at a frequency proportional to its energy E_n. Therefore tracking the global time evolution of the wavepacket simplifies to summing these independently rotating components.

In practice, evaluating the entire spectrum up to dimension D is computationally prohibitive. Therefore, a basis truncation is applied at a finite number of requested eigenstates $n_{\text{ev}} \ll D$. The truncated state propagation is valid provided the initial wavepacket's energy density is contained well within the resolved energy window, this is verified at runtime via the projection completeness condition $\sum_{n=1}^{n_{\text{ev}}} |c_n|^2 \geq 0.95$.

2 Methods

The numerical framework is implemented in Python, utilizing DOLFINx for the automated assembly of the variational forms [2]. The linear algebra backends are managed via the Portable, Extensible Toolkit for

Scientific Computation (PETSc)[3] and the Scalable Library for Eigenvalue Problem Computations (SLEPc) [7].

2.1 Mesh

There are three main meshes currently implemented. A rectangular domain, a circular, and a non-integrable chaotic Bunimovich stadium. All three are implemented using Gmsh [5], which are subsequently converted into native `DOLFINx` structural objects via `dolfinx.io.gmshio.model_to_mesh`. Across all configurations, boundaries are synchronized with the geometric kernel and grouped to allow uniform downstream enforcement of homogeneous Dirichlet boundary conditions.
The square domain is defined across $[-L/2, L/2]^2$. It is generated via the OpenCASCADE kernel using the function `model.occ.addRectangle`. The circular domain is generated via the function `model.occ.addDisk`. The horizontal and vertical radius parameters are set identically to a single value R. Because the boundary exhibits a non-orthogonal curvature, the characteristic element sizes are explicitly constrained by a user-specified spatial sizing parameter $h = h_{\text{target}}$ to regulate Delaunay refinement density along the circumference.The Bunimovich quantum stadium domain combines curved and flat boundaries to break spectral integrability. It is created using a constructive solid geometry (CSG) using three separate spatial primitives: a central rectangle primitive of length L_{straight} and height $2R$, positioned from $x_{\min} = -L_{\text{straight}}/2$ to $x_{\max} = L_{\text{straight}}/2$, and from $y_{\min} = -R$ to $y_{\max} = R$, a left circular disk primitive of radius R centered at the left focal coordinate $(-L_{\text{straight}}/2, 0, 0)$ and a right circular disk primitive of radius R centered at the right focal coordinate $(L_{\text{straight}}/2, 0, 0)$. These three intersecting independent surfaces are structurally combined into a continuous two-dimensional domain surface using the `model.occ.fuse` operator. A subsequent execution of `model.occ.synchronize` purges the overlapping internal boundary curves, leaving a single exterior bounding envelope wrapping the stadium topology.

2.2 Solver

The solver receives a mesh and then formulates the continuous weak form, enforces homogeneous Dirichlet boundary conditions, and handles the global sparse matrix assembly. The actual eigenvalue problem is solved using SLEPc and the eigenvectors and energies are returned. These can then be used to solve time evolution via spectral time propagation.

2.2.1 Spectral Time Propagation

In order to project the initial real-valued wavepacket ψ_0 onto the eigenbasis, it is first evaluated on the spatial coordinate vertices of the finite element triangulation $\mathcal{T}_h$. To preserve probability density conservation across time, the initial vector is normalized. This is executed by computing the matrix-vector product $M\boldsymbol{\psi}_0$ via `M.mult`, evaluating the complex dot product `np.vdot`, and scaling the base array $\boldsymbol{\psi}_0 \leftarrow \boldsymbol{\psi}_0/\|\psi_0\|_M$. The normalized coordinates are then projected onto the resolved spatial eigenbasis matrix $\boldsymbol{\Phi} = [\boldsymbol{\phi}_1, \ldots, \boldsymbol{\phi}_{n_{\text{ev}}}]$ to yield the spectral distribution coefficients $c_n \in \mathbb{C}$ necessary for the simplified time evolution:

$$\mathbf{c} = \boldsymbol{\Phi}^\dagger M \boldsymbol{\psi}_0. \tag{10}$$

The time domain is discretized into uniform steps based on the users input. At each discrete time snapshot t_k, the continuous wavepacket is reconstructed by evaluating the time-evolution operator:

$$\boldsymbol{\psi}(t_k) = \sum_{n=1}^{n_{\text{ev}}} c_n e^{-iE_n t_k} \boldsymbol{\phi}_n. \tag{11}$$

This operation is done using vector-matrix operations: `(coeffs * np.exp(-1j * energies * t)) @ eigvecs`.

2.2.2 Wavepackets

For the scenario studies, spatial disturbances are introduced into the domain in the form of localized quantum wavepackets. Currently using up to four wavepackets is implemented. Each wavepacket configuration requires

specifying its spatial center coordinates (x_0, y_0), its characteristic spatial width σ, and its directional initial momentum components (k_x, k_y). Using these parameters, the initial global state $\psi_0(\boldsymbol{x})$ is constructed as a superposition of moving Gaussian wavepackets, where an individual packet is mathematically formulated as:

$$\psi_0(\boldsymbol{x}) = \exp\left(-\frac{(x-x_0)^2 + (y-y_0)^2}{2\sigma^2}\right) \exp\Big(i(k_x x + k_y y)\Big). \tag{12}$$

When selecting parameters for the initial state, a careful balance must be maintained between real-space localization (σ) and momentum-space band-limiting (k). According to the Heisenberg Uncertainty Principle ($\Delta x \cdot \Delta k \geq 1/2$), a small σ forces the particle into a small region. That means that a massive range of high-energy components are injected into the initial state. In order for the solver to handle this accurately over time a large number of high-energy eigen states is required. So choosing the right parameters in regards to the number of eigenstates and mesh size is important here.

2.3 Lieb-Robinson

The Lieb-Robinson bound is calculated and visualized by first initializing a wavepacket as described above. This wavepacket is then evolved in time and the wave spreads are tracked. For every timestep the code scans outward from the center and finds the coordinates where the probability density drops below a threshold, here 10^{-4}. This boundary is considered the farthest the wave has reached. By running a linear regression using `np.polyfit` on how the farthest lines expand over time the experimental Lieb-Robinson speed limit can be computed. This can then be compared to the analytical velocity bound, which is calculated based on the assumption that maximum possible speed of the system is based on the highest resolved energy level. Finally the Lieb-Robinson exponential tail or leakage tail is investigated. For this towards the end of the simulation a snapshot is chosen that looks outside the wave boundary and fits a line to the logarithm of the density. The resulting slope, according to the Lieb-Robinson theorem, must be an exponential decay constant.

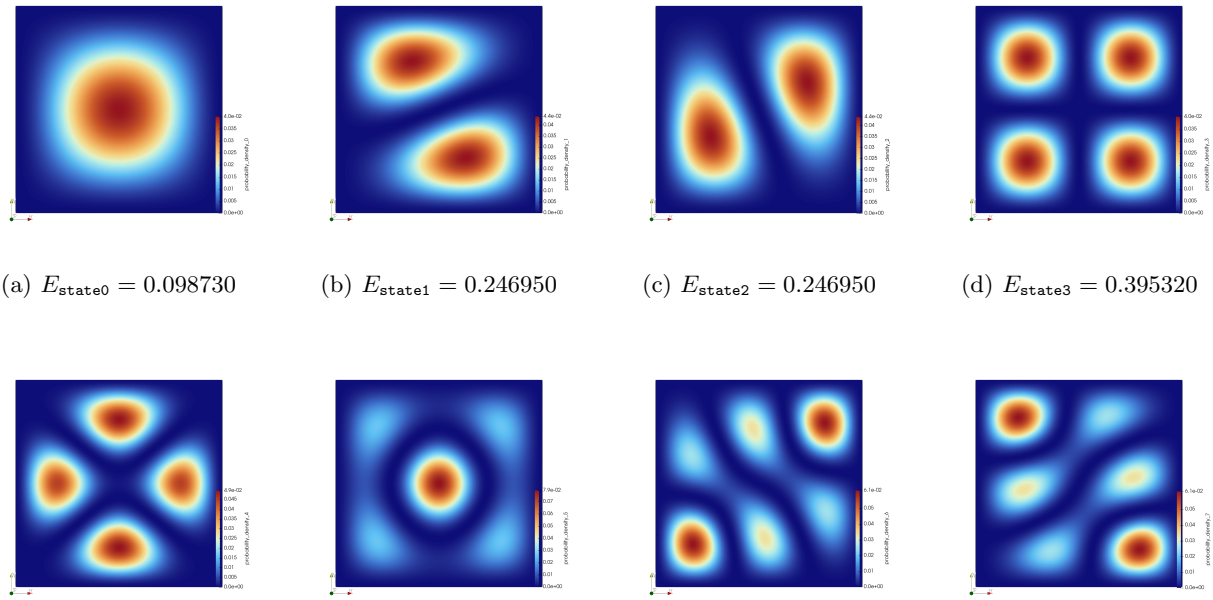
2.4 Plotting and Data Visualization

For postprocessing and visualization, a mix of Python plotting and ParaView [1] are used. `Matplotlib`[8] is used to generate diagnostic charts, space-time heatmaps, and spatial slice profiles. The grid geometry and the calculated wave functions are written to a localized `/results` directory. Static spatial data, such as calculated energy eigenstates, are stored using the `XDMF` and `HDF5` file formats via FEniCSx (`dolfinx.io`). For the time-evolution data, `PVD` file containers are saved to capture successive temporal frames efficiently. For the reproducibility of any figures used that are not automatically generated via `Matplotlib`, the corresponding ParaView statefiles (`.pvsm`) are preserved.

3 Results and Discussion

For verification and qualitative validation, a baseline benchmark test case of a square domain with side length $L = 10.0$ and grid refinement $N = 60$ was evaluated. The physical probability densities $|\psi_i|^2$ of the first eight resolved quantum eigenstates are visualized in a 2×4 grid configuration in Figure 1.

The computed distributions demonstrate strict adherence to the expected geometric symmetries of the C_{4v} point group. Panels (a) through (d) cleanly isolate the lowest-frequency analytical modes and their degenerate partners. Crucially, the squaring of the complex wavefunction amplitudes maps all structural phase regions to strictly non-negative values, leaving crisp, distinct nodal lines (zero-probability corridors) visible as deep blue paths separating the localization peaks. Panels (e) through (h) illustrate the intricate lattice structures resulting from linear combinations within the higher-order degenerate energy subspaces resolved by the parallel SLEPc eigensolver, serving as a robust visual confirmation of the spatial fidelity of the finite element discretization.



(a) $E_{\texttt{state0}} = 0.098730$ (b) $E_{\texttt{state1}} = 0.246950$ (c) $E_{\texttt{state2}} = 0.246950$ (d) $E_{\texttt{state3}} = 0.395320$

(e) $E_{\texttt{state4}} = 0.494323$ (f) $E_{\texttt{state5}} = 0.494323$ (g) $E_{\texttt{state6}} = 0.642942$ (h) $E_{\texttt{state7}} = 0.642943$

Figure 1: Spatial probability density distributions $|\psi_i|^2$ for the first eight resolved energy eigenstates ($i \in \{0, \ldots, 7\}$) within an infinite 2D square potential well. The dark channels map out structural quantum nodes where the localization probability is zero, capturing the geometric symmetries of the domain.

3.1 Verification

The verification is done for the simple test case of a square domain with L = 10.0 and N = 60 for 20 modes. Further the verification is split into two parts, the eigenvalue solver verification and the time evolution verification.

For the solver verification the analytical solution, $E_{nx,ny} = \pi^2/2 * (nx^2 + ny^2)/L^2$, is compared to the numerical solution for each mode. As can be seen in Figure 2 the analytical and FEM solutions are identical and the relative error stays below 10^{-2}. This verifies the accuracy of the solver.

In order to test the time evolution a stationary state validation test is run. For this a trivial physical scenario is chosen. The numerical ground state computed is used as the initial state and propagated through time. From a quantum mechanics perspective this energy eigenstate $\boldsymbol{\phi}_n(\boldsymbol{x})$ is called a stationary state because the probability density $|\boldsymbol{\psi}_{\texttt{exact}}|^2$ is completely time-invariant. Therefore the time-dependent solution simplifies to a quantum phase rotation as a function of time.

$$\boldsymbol{\psi}_{\texttt{exact}}(\boldsymbol{x}, t) = e^{-iE_n t}\boldsymbol{\phi}_n(\boldsymbol{x}). \tag{13}$$

This means the exact analytical solution is a simple complex exponential and allows for direct comparison between the analytical and the FEM solution. This is visualized in the left panel of 3, which shows the time evolution of the ground-state wave oscillations. The FEM solution is identical to the analytical solution confirming that the solver handles quantum phase and amplitude correctly over time. The right panel in Figure 3 shows the unitary norm conservation error: $\texttt{error} = |\langle\psi_{\texttt{num}}| \boldsymbol{M} |\psi_{\texttt{num}}\rangle - 1.0|$, this checks that the total probability must remain perfectly conserved at 1 across all time. It also shows the coherent state overlap fidelity error: $\texttt{error} = |\langle\psi_{\texttt{exact}}| \boldsymbol{M} |\psi_{\texttt{num}}\rangle - 1.0|$, this measures the phase and alignment error based on the projection magnitude between the numerical solution and the exact solution. If the overlap falls below 1.0

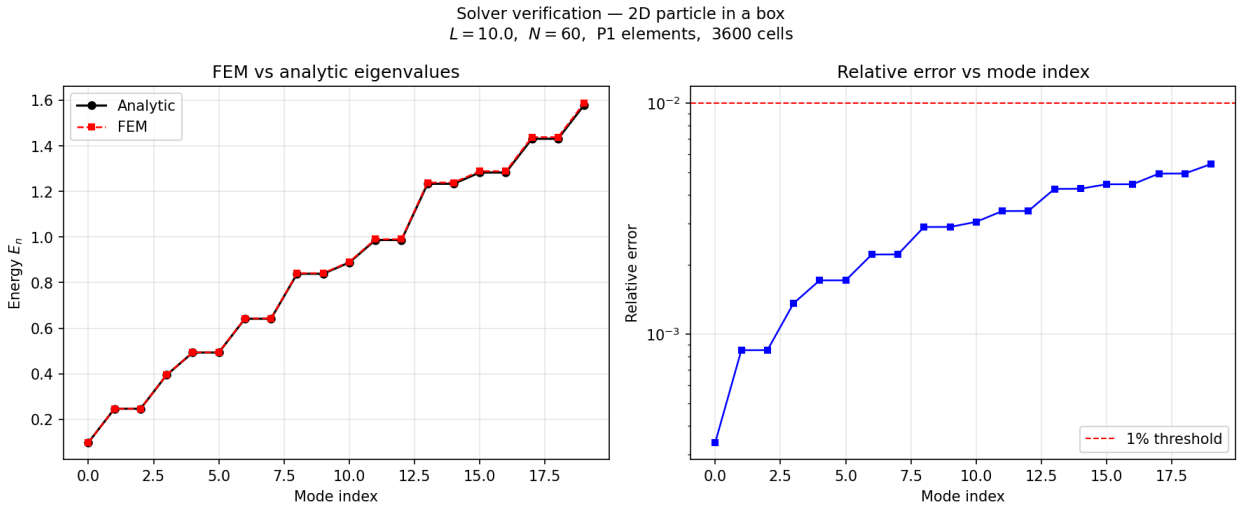


Figure 2: Solver verification

there is an issue. As can be seen in Figure 3 both errors remain at 10^{-16} across the entire simulation since no numerical time-stepping method is used. This confirms that the time evolution is working as intended.

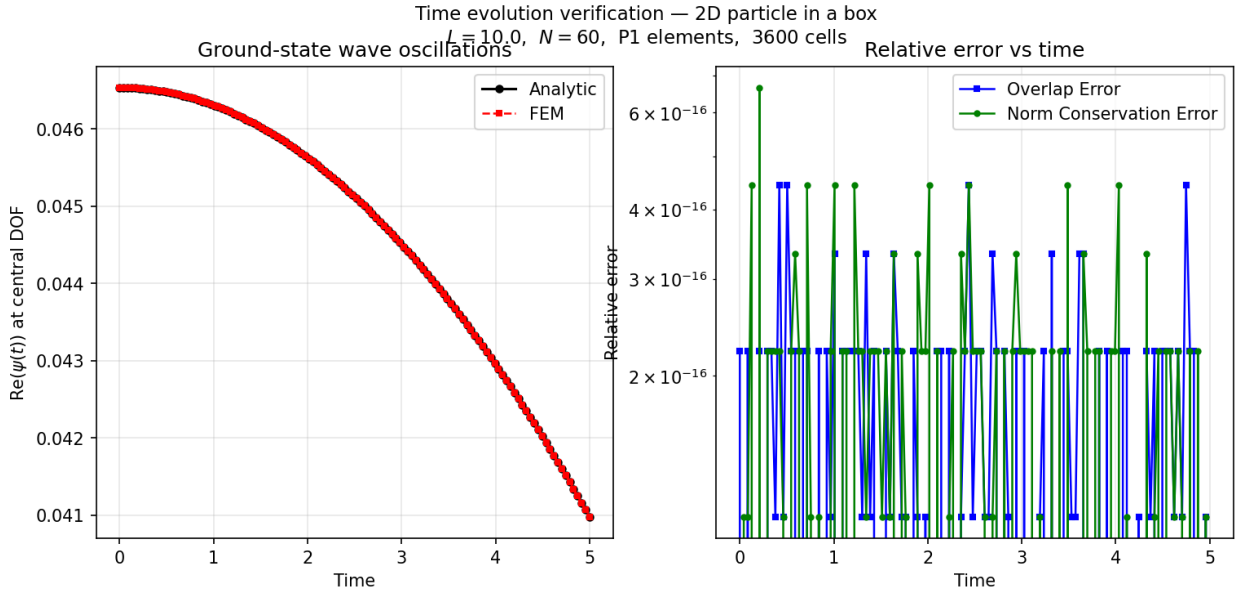


Figure 3: Time evolution verification

Combining both verification schemes, the solver clearly captures the underlying physics of the problem. To confirm its numerical stability and accuracy, a spatial mesh convergence study was performed.

3.2 Mesh convergence study

The mesh convergence study was also performed for a square domain with ($L = 10.0$) across a sequence of progressively refined grids with $N \in \{8, 16, 32, 64, 96\}$, tracking the errors of the first 10 solved energy

eigenstates. Two norms were calculated:the global L^2-norm error, defined as

$$\|\psi_{\text{exact}} - \psi_{\text{num}}\|_{L^2} = \left(\int_\Omega |\psi_{\text{exact}}(\mathbf{x}) - \psi_{\text{num}}(\mathbf{x})|^2 \, d\Omega \right)^{1/2}, \tag{14}$$

and the localized maximum (or L^∞) norm error, defined as

$$\|\psi_{\text{exact}} - \psi_{\text{num}}\|_{L^\infty} = \max_{\mathbf{x}\in\Omega} |\psi_{\text{exact}}(\mathbf{x}) - \psi_{\text{num}}(\mathbf{x})|. \tag{15}$$

The resulting error curves are displayed on a log-log scale in Figure 4. A reduction in mesh size $h = L/N$ yields a monotonic decrease in both error metrics. The convergence rates match the theoretical asymptotic limits for linear P1 elements $\mathcal{O}(h^2)$. The Root-Mean-Square (RMS) error and the maximum error converge at rates of approximately 2.00 and 2.01, respectively, exhibiting excellent alignment with the ideal quadratic reference slope. This validates the numerical stability of the solver.

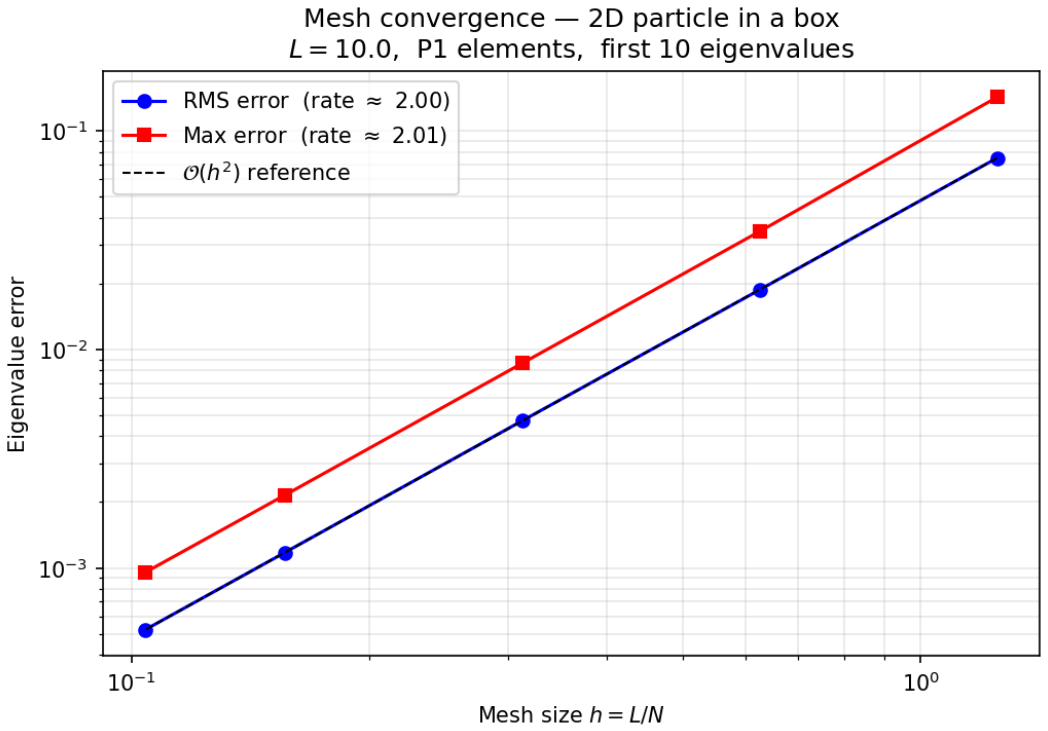


Figure 4: Eigenvalue error convergence as a function of the characteristic mesh size $h = L/N$ for the first 10 modes, plotted alongside an $\mathcal{O}(h^2)$ theoretical reference line.

3.3 Realistic scenario study

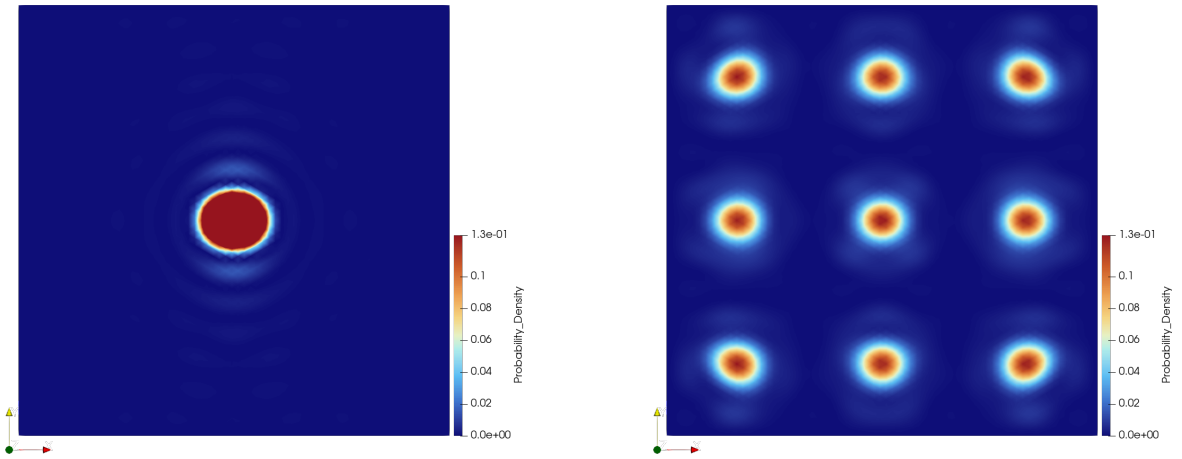
For this scenario study a disturbance is introduced into the domain. This a highly localized "blip" of quantum information at t=0. A few cases were explored. Firstly the basic 2D square case. This case is also used to explore the concept of Lieb Robinson bounds. Secondly the circular geometry and finally a quantum stadium.

3.3.1 Square Box Geometry

For the square box geometry, a domain with side length $L = 10.0$ and grid density $N = 60$ was simulated using $n_{\text{ev}} = 150$ resolved energy modes. A single wavepacket ($n_{\text{packets}} = 1$) is initialized at the exact center

of the domain $(x_0, y_0) = (0.0, 0.0)$ with an initial width of $\sigma = 0.4$ and momentum components $k_x = 2.0$ and $k_y = 2.0$. The time evolution was spectrally propagated from $t = 0$ to $t = T_{\max} = 5.0$, saving 120 uniform temporal snapshots for analysis.

The isotropic expansion of the wavepacket from the center results in highly symmetric reflections once the wavefronts reach the orthogonal, hard-wall homogeneous Dirichlet boundaries. Through the subsequent constructive and destructive self-interference of these scattered wave components, a regular, checkerboard-like grid pattern emerges within the probability density profile. These initial and final states are visualized in Figure 5. The corresponding full time-evolution video can be found in the project repository at /meshes.md.



(a) Square initial state ($t = 0$)

(b) Square scattered state ($t = 5$)

Figure 5: Probability density snapshots in the square box domain at $t = 0$ and $t = 5$.

3.3.2 Lieb-Robinson Bounds Quantification

To numerically verify the Lieb-Robinson bounds, a scenario similar to the square box is configured, but with an asymmetric initial wavepacket placement at $x_0 = -L/4 = -2.5$. Pushing the initialization center toward the left wall provides a long, open propagation corridor toward the right boundary. This spatial asymmetry allows the tracking algorithm to map the unobstructed, ballistic expansion of the leading edge over a longer period before boundary reflection waves eventually contaminate the profile.

To process the two-dimensional probability density data for linear tracking, a one-dimensional central slice is taken along the axis of propagation for rectangular domains, while an annular or concentric angular integration ($\langle |\psi|^2 \rangle_\theta$) is applied to radial geometries. The spatial tracking of the information front over time is governed by a visibility threshold mask set to 10^{-4} of the peak density. The coordinates where the wave magnitude meets this threshold define the dynamic frontier of the information light cone. By applying a first-order linear polynomial fit (`np.polyfit`) to these frontier coordinates over time, the numerical group velocity ($v_{\text{LR}} = 0.223$) is extracted and overlaid against the analytical upper bound ($v_{\text{analytic}} = 11.85$).

Figure 6 visualizes a comparative analysis of the space-time heatmaps with and without the analytical and experimental light cone boundaries. Furthermore, the exponential leakage tail extending into the classically forbidden region outside the light cone is evaluated. As shown in the comprehensive diagnostic breakdown in Figure 7, a linear regression applied to the logarithm of the probability density confirms that any superluminal

information leaking past the nominal wavefront decays strictly at an exponential rate ($\psi \sim e^{-\mu x_{\text{rel}}}$), yielding a spatial decay constant of $\mu = 1.43$. This result rigorously demonstrates the bounded nature of localized quantum state propagation inside finite element meshes.

3.3.3 Circular Geometry

Next, a circular geometry is simulated to observe isotropic wave propagation and boundary scattering. The domain is defined with a radius of $R = 5.0$ and discretized with an unstructured mesh using a target sizing parameter of $h_{\text{target}} = 0.2$. To capture the radial dynamics, a single localized wavepacket ($n_{\text{packets}} = 1$) is initialized directly at the origin $(x_0, y_0) = (0.0, 0.0)$ with an initial width of $\sigma = 0.4$ and directional momentum components of $k_x = 2.0$ and $k_y = 2.0$.

The system is spectrally propagated over a total time horizon of $T = 5$. As the wavepacket expands radially outward, it hits the curved boundaries of the circular cavity. The symmetric reflection back toward the origin causes the wave components to collide with themselves, yielding clean, concentric interference rings that wrap around the center of the domain. These probability density profiles at the initial and final states are visualized in Figure 8. The corresponding full time-evolution video can be found in the project repository at /meshes.md.

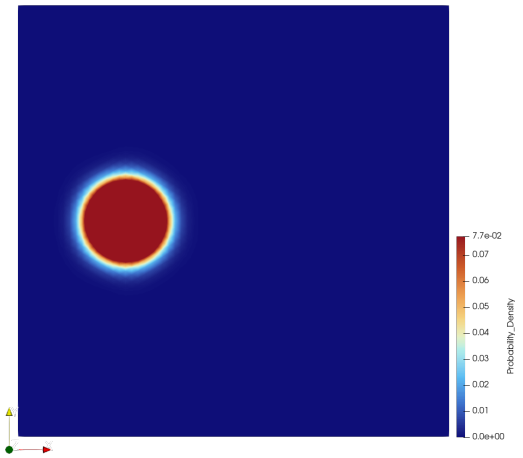
3.3.4 Quantum Stadium

Finally, a quantum stadium geometry is simulated to study chaotic wave propagation. It is worth noting that while this domain shares the geometric silhouette of a Bunimovich stadium, it serves as a preliminary test case rather than a strict realization of the classic non-integrable system, yet it remains a highly interesting scenario to observe.

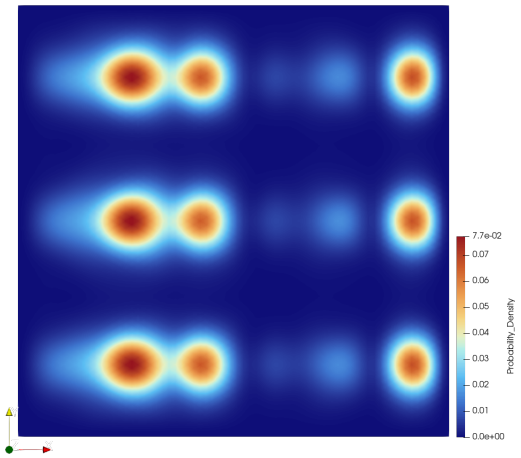
The domain consists of a central rectangle with a straight length of $L_{\text{straight}} = 10.0$ and semicircular caps with a radius of $R = 5.0$, discretized using an unstructured grid with a target mesh sizing parameter of $h_{\text{target}} = 0.2$. The underlying generalized eigenvalue problem was solved to extract the first $n_{\text{ev}} = 150$ lowest energy eigenstates.

The initial state was modeled as a symmetric superposition of two colliding wavepackets ($n_{\text{packets}} = 2$) initialized along the centerline ($y_0 = 0.0$). The left packet was positioned at $x_0 = -2.5$ with a positive horizontal momentum kick of $k_x = +10.0$, while the right packet was mirrored at $x_0 = +2.5$ with a negative momentum kick of $k_x = -10.0$. Both wavepackets were assigned an identical spatial width parameter of $\sigma = 0.2$ and zero vertical momentum ($k_y = 0.0$). The system was then spectrally propagated over a total simulation time horizon of $T = 10.0$, with the resulting probability densities serialized across 500 uniform temporal frames into a master `.pvd` index for post-processing and animation within ParaView.

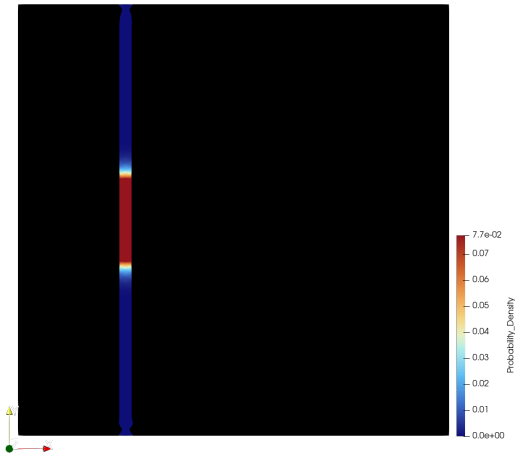
As the time-dependent state progresses, the waves propagate outward and encounter the curved semicircular boundaries. These geometries break the regular symmetry of the wavefunctions, inducing complex boundary reflections and chaotic scattering. This rapid wave mixing quickly dissolves the highly localized initial packets into an intricate, distributed quantum interference pattern across the cavity. The spatial profile snapshots at the initial and final states are visualized in Figure 9, and the full time-evolution video is hosted in the project repository at /meshes.md.



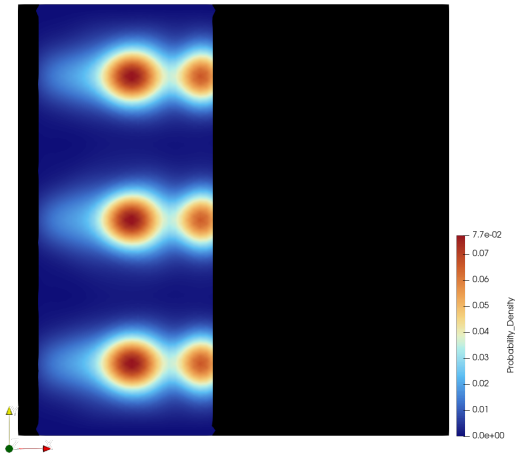
(a) Space-time density without cone



(b) Space-time density without cone (f_1)



(c) Light cone tracking limits



(d) Light cone tracking limits (f_1)

Figure 6: Space-time probability distributions comparing unmapped profiles against explicit analytical and experimental light cone boundaries.

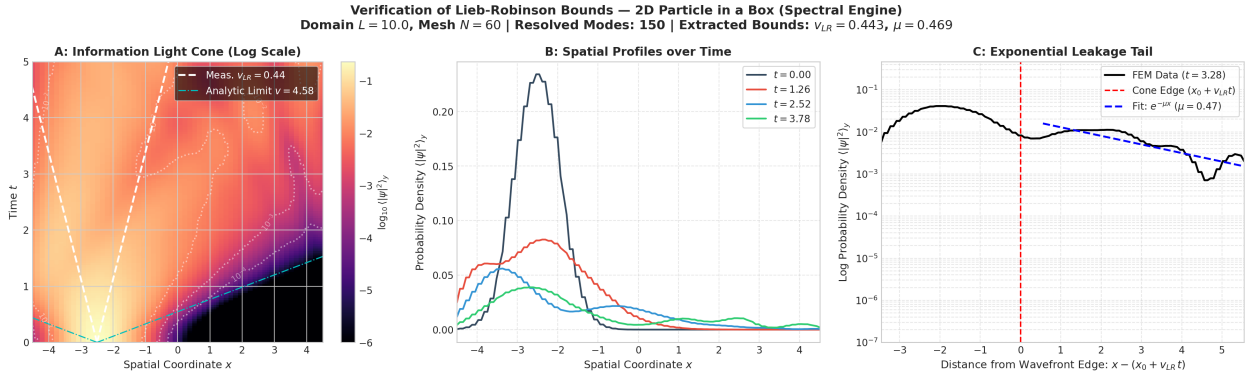
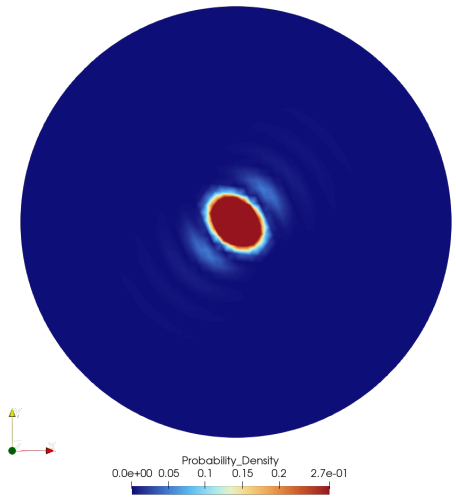
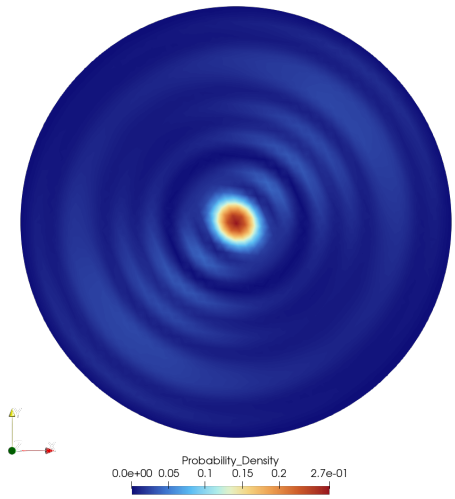


Figure 7: Verification of Lieb-Robinson bounds showing (A) the space-time information light cone, (B) spatial profile slices over time, and (C) the exponential leakage decay profile fit outside the wavefront edge.

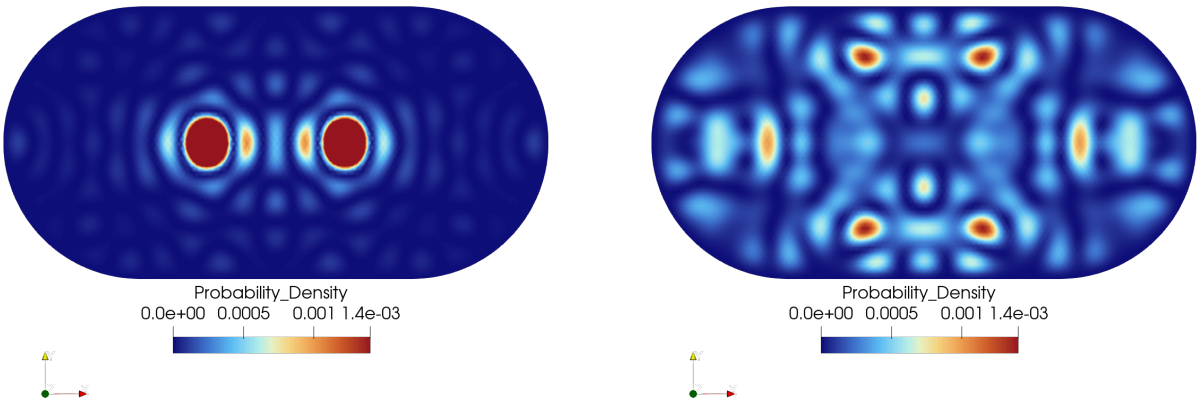


(a) Circle initial state ($t = 0$)



(b) Circle scattered state ($t = 5$)

Figure 8: Probability density snapshots in the circular domain at $t = 0$ and $t = 5$.



(a) Stadium initial state ($t = 0$)

(b) Stadium scattered state ($t = 5$)

Figure 9: Probability density snapshots in the stadium domain at $t = 0$ and $t = 5$.

4 Conclusion

A generalized eigenvalue solver was successfully developed and implemented using the DOLFINx finite element framework, coupled with an analytical spectral time-evolution scheme. The code was verified for a simple square. A mesh convergence study was performed, which clearly demonstrated the expected spatial convergence behavior and validated the accuracy of the finite element discretization.

By propagating a localized quantum wavepacket at $t = 0$ within the square domain, the system's dynamic evolution was observed. The simulation highlighted features reminiscent of a Lieb–Robinson-type bound, illustrating that quantum information propagates at a finite effective velocity even within a continuous, non-relativistic quantum system. However, it is crucial to emphasize that these observations do not constitute a definitive or rigorous proof of a fundamental quantum principle.

Further work for this project could include implementing alternative temporal discretizations and / or integrating a qiskit-based quantum simulation pipeline to allow for a direct comparison between the classical finite element wave dynamics and a quantum simulation.

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